

Swesan Pathmanathan

Canadian Citizen

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EDUCATION

McMaster University

Bachelor of Engineering, Software Engineering (Co-op)

Hamilton, ON

Sept 2021 – Apr 2026

TECHNICAL SKILLS

Languages: Python, C++, C, TypeScript, SQL, Java

Hardware IP Domains: RAS, Int-IO, Ext-IO, GFX, PPA, Memory, System Management

Databases: PostgreSQL, Supabase

AI Tools: Cursor, GitHub Copilot

Frameworks & Libraries: React Native, Node.js, Next.js, React

AI & ML Libraries: PyTorch, TensorFlow/Keras, Scikit-learn, NumPy, Pandas, Matplotlib, rocJPEG

Systems & Infrastructure: Linux, Ubuntu, RHEL, CentOS, Windows, KVM, QEMU, Docker, GitHub Actions, ROCm, BMC, BIOS/UEFI, Vercel

Developer Tools: Git, Bash, Quartus

WORK EXPERIENCE

Advanced Micro Devices (AMD) | Python, C/C++, Linux, Ubuntu, RHEL, CentOS, BKC, ROCm, KVM, QEMU, Docker, Git, Bash, Windows, BMC, BIOS/UEFI, RAS, Int-IO, Ext-IO, GFX, PPA, Memory, System Management

Markham, ON

Datacenter GPU System Integration Engineer (Co-op)

May 2024 – Aug 2025

- Rotated as **test lead** on MI300X, MI325X, and MI350X post-silicon lanes; tracked **100+ diagnostic runs weekly** on **Linux** and **Windows** stations across **RAS, Int-IO, Ext-IO, GFX, PPA, Memory, and System Management**.
- Updated **Linux** GPU racks on **Ubuntu, RHEL, and CentOS** and **Windows** systems to validated **BKC** baselines; flashed **BMC** and **BIOS/UEFI**, staged **Docker** images, and configured test configs and ML workload parameters so **NPI**-partitioned workloads could run on **MI350X**.
- Set up **KVM/QEMU** on **MI325X CentOS** hosts, wrote setup docs the team reused for running workloads, validated **MI325X KVM** guests under production-style workloads, and triaged failures found during those runs.
- Debugged **BMC** and **BIOS/UEFI** failures on MI300X, MI325X, and MI350X racks from mixed customer configs; wrote flash and rack guides reused on **20+** bring-ups, **cutting repeat setup questions by about 40%**.
- Built **Python** and **Bash** scripts plus an internal workload launcher that cut test launch time from **about 20 minutes to under 5 minutes** per run for the validation team.
- Triaged defects across **ROCm** code and analyzed **C/C++** platform code for root cause; documented debug and repro steps so similar issues resolved faster, **cutting reopened defects on my queue by about 30%**.
- Ran **rocJPEG** workloads on MI systems and debugged **GFX** IP validation failures on the graphics and multimedia path during post-silicon passes.

PROJECTS

Hitchly Ride Share Application | Node.js, PostgreSQL, Docker

Sept 2025 – Apr 2026

- Implemented CI and release checks with GitHub Actions and **Docker** for linting, tests, and pull request validation, strengthening production software quality gates for each backend change.
- Built and integrated backend **APIs** in **Node.js** with a **Docker**-containerized **PostgreSQL** service, supporting client-server workflows and feature integration across the application stack.
- Integrated mobile trip-booking flows with backend APIs across **React Native, TypeScript, and Node.js** services, improving end-to-end feature delivery across client and server components.

Gym App Platform | Next.js, React, TypeScript, Supabase, PostgreSQL

May 2026

- Built a full-stack gym tracking platform with **Next.js** App Router and **React**, using **TypeScript** for shared client/server types across workout, history, and progress flows.
- Implemented backend data access with **Supabase** and **PostgreSQL**, including SQL migration workflows for schema evolution and reproducible environment setup.

- Developed analytics views with **Recharts** and date-based trend rollups with **date-fns**, and prepared deployment-ready configuration targeting **Vercel** with basic PWA add-to-home-screen support.

Linux Log Normalization and Token Frequency Analysis | Python, NumPy, pandas

Jan 2026

- Developed a configurable text normalization pipeline in **Python** for processing raw **Linux logs**, supporting tokenization, stemming, lemmatization, and digit filtering.
- Performed token frequency computation and statistical aggregation using **NumPy** and **pandas** to analyze distributional patterns in **Linux logs**.

Ontario Hospital Association Database Design | SQL

Sept 2025

- Designed a relational schema for a hospital information system modeling patients, records, employees, and clinical workflows to support consistent operational queries.
- Authored **SQL** scripts implementing primary keys, foreign keys, and referential integrity constraints to ensure consistency across patient and record data.
- Developed a full database creation and deployment script with normalized relational schemas for reproducible **SQL**-based rollout.

SufSort Sorting System | C, Bash, Linux

Nov 2025

- Wrote two sorting programs in **C** to compare how different algorithms behave on large data files of up to several hundred megabytes.
- Used **Bash** scripts on **Linux** to generate input data, run batch tests, and check sorted output, catching logic errors early in development.
- Measured runtime behavior across multiple input sizes and used test logs to compare sorting throughput, making algorithm tradeoffs visible for larger file workloads.

Linux Log Corpus Analysis | Python, scikit-learn, NumPy

Feb 2026

- Built a simple analysis pipeline in **Python** to read and process tens of thousands of labeled Linux system logs.
- Used **NumPy** and **scikit-learn** to find which log terms best separate different log categories, improving signal-to-noise for downstream analysis.

Pneumonia Detection CNN | Python, TensorFlow/Keras, NumPy, pandas, scikit-learn

Jan 2026

- Built a convolutional neural network classifier in **Python** using **TensorFlow/Keras** to detect pneumonia from chest X-ray images.
- Applied data preprocessing, dataset splitting, and augmentation using **NumPy** and **pandas** to improve model generalization on imbalanced clinical datasets.
- Evaluated model performance using **accuracy**, **precision**, **recall**, confusion matrices, and visualization tools to guide architecture and hyperparameter tuning.

Breast Cancer Prediction Neural Network | Python, PyTorch, NumPy, scikit-learn

Sept 2025 – Dec 2025

- Built a feedforward neural network with **PyTorch**, using **NumPy** and **scikit-learn** preprocessing and stratified splits, to classify malignant and benign breast cancer samples from tabular clinical datasets.
- Applied regularization techniques including **dropout** and **L1/L2** penalties to improve model stability and reduce overfitting.

ALU Design with Sequential Circuits | Verilog, Quartus

Sept 2025

- Structured a synthesizable ALU datapath from small **Verilog** blocks and checked behavior block-by-block before full-chip stitching.
- Coded deterministic **Verilog testbench** stimulus plus self-checks, exercised ALU arithmetic and logic modes over a scripted input matrix, and used **Quartus** waveform debug until every planned functional regression passed.

Terrain Generation System | Java, Maven, Git, JTS

2024

- Developed a software tool in **Java** for generating and visualizing 2D terrain meshes for simulation and visualization workflows.
- Modeled vertices, edges, and polygons using the **JTS** geometry library to represent terrain features and topology.
- Implemented a command-line interface to configure generation parameters, run batch experiments, and visualize output meshes.

RELEVANT COURSES

Performance Analysis Tools and Methods, Concurrent Systems and Synchronization, Computer Networks Security, Database Systems

HOBBIES & INTERESTS

Gaming, Hardware/Software Tinkering, Gym.